Shared evidence | Source and town data

Disease data newsroom: make the numbers fair

Name ____________________________ Date _______________

The WHO card is a real source summary. The town dataset is constructed for this lesson; keep them separate.

WHO source card (paraphrase): the page dated 25 September 2025 reports at least 43 million deaths from non-communicable diseases worldwide in 2021. These were 75% of global deaths unrelated to the pandemic. The measure is deaths, not infections. Source: WHO, Noncommunicable diseases; checked 8 September 2026.

Source link: https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases

Town	New cases in 7 days	People at risk at start
A	24	3000
B	18	1000
C	30	6000

Constructed Town Bulletin: three fictional towns; same infection definition, same 7-day period, no existing cases at start; count each new case once. Assume comparable recording. Age/exposure differences are not supplied.

W1. Calculate per 1000. W2. Compare count and proportion. W3. Improve the unsupported water-cause claim.

Supported route | Make the fair comparison

Formula: new cases ÷ people at risk ×1000. Use the shared page. Complete a paper infographic below.

S1. Calculate cases per 1000 for A, B and C. Show one calculation.

S2. Sketch three bars from zero, label towns and “new cases per 1000 people over 7 days”.

S3. Add a headline naming the town with the highest proportion. Add “constructed data” and one thing the table cannot prove.

S4. For the WHO source, record the data year and what the 75% denominator is. Alternatively audit the Town Bulletin’s period and population.

Standard route | Create the infographic

Use the space as a layout plan, then finish in a document/slide if available.

M1. Calculate each town's cases per 1000, showing one method.

M2. Draw an infographic with a zero-based bar chart, labelled units/period and a factual headline.

M3. Add a constructed-data source note and a limitation that prevents a causal claim.

M4. Audit the WHO source: publisher, publication date, data year, measure and 75% denominator. Or audit the Town Bulletin's equivalent details.

Stretch route | Audit an editor's claims

Optional data challenge. Same constructed 7-day period: town D records 6 new cases among 200 people; town E records 30 among 2000.

T1. Calculate D and E per 1000. Which has the higher count and which the higher observed proportion?

T2. In town D one extra recorded case would change its value by how much per 1000? Explain why a small count deserves caution.

T3. Can the original town data prove that a prevention programme caused B's value? Give two missing pieces of evidence or study-design issues.

T4. Draft a two-sentence accurate infographic caption for D/E, including constructed-data status, period and a limit.

Exit and reflection | Numbers need context

A clear label is part of the science.

E 1. Calculate 10 new cases among 2000 people per 1000 for one stated 7-day period.

E 2. Why distinguish the publication date from the data year?

E 3. Why can a town comparison not by itself prove a cause?

L1. What did your editor ask? Improve one label or explain why it was accurate.

Next useful step

A. Keep the population denominator.

B. Show units and period.

C. Check the original source.

D. Limit the causal claim.